Figure 3. Size-dependent susceptibility links deterministic perturbations to weak-input domain drift

A. Exact preceding Wave-pinning activity profiles, C=30, for a cue straddling the incumbent edge (i) or separated from it (ii). Rows are initial, half-turn, 90% turn and cue-free relaxation; the grey band is cue support, diamonds are saddles and the black tick is heading. All A data and timings are unchanged.

B. Exact preceding half-turn time against cue-edge gap for the three matched sizes, light to dark: C=30,37.8,47.6; Spring 2.5, 3.15, 3.97. Lines mark first saddle appearance; grey marks undefined heading at one or more sizes. Open upward triangles and arrows denote lower bounds. Continuous-cue data and native parameters are unchanged.

C. A weak input anywhere shifts the domain by g(x) times dose; noise is many such inputs; drift follows from g and the input covariance; nothing is fitted. Lines are the terminal displacement per dose predicted by the linearized native model for the standard Gaussian pulse centred at each arc position: SD 1.5, duration 50 (Spring 0.15, 5), at f=0.001. Light and dark are C20/C60 (Spring 1/3). New nominal-amplitude tests from settled states are filled circles; the existing edge pulse is an open square. Both subtract a matched unpulsed control. The pulse confirms g at the edge; the map elsewhere comes from the linearized native model and is tested at domain centre, halfway to the edge, edge±1/3 pulse SD, quarter-basal arc and antipode. Maximum absolute gain error normalized to map amplitude is 1.69%. Terminal edge gain large/small: Wave-pinning 0.701, Holmes 0.914, Spring 0.989, Goryachev 0.088.

D. Mean-state activity in light grey on its own axis, sharing the centred physical arc coordinate with C and E. Dotted verticals locate half-contrast edges. Spring positions are material fractions times the mean-state circumference.

E. Signed per-position contribution c_j/D_s at the same sizes and input, using raw native g and the unchanged noise covariance, with no measured-gain rescaling. Contributions sum to one; negative values are retained. Edge zones are within two input correlation lengths of the nearest edge; plateau interior and basal arc exclude these zones. Fractions of D_s, left/right edge, plateau interior, basal arc respectively: Wave-pinning C20: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000; Wave-pinning C60: left edge 0.499, right edge 0.499, plateau interior 0.001, basal arc 0.000; Holmes C20: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000; Holmes C60: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000; Spring C1: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000; Spring C3: left edge 0.498, right edge 0.498, plateau interior 0.002, basal arc 0.002; Goryachev C20: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000; Goryachev C60: left edge 0.500, right edge 0.500, plateau interior 0.000, basal arc 0.000. Raw-native and measurement-normalized totals are both exported.

F. Wave-pinning C20 and C60: pulsed falling edge (solid) and opposite rising edge (dashed) move together and retain a permanent offset after the grey pulse-on interval. Displacement is divided by pulse dose. Width and centroid histories, other models, every amplitude and duration are supplementary.

Figure 3. Size-dependent susceptibility links deterministic perturbations to weak-input domain drift · continued

G. Predicted against measured physical domain diffusion after each is divided by its own fixed-smallest-size value at the same input. Only the shared-translation prediction appears here; the grey line is equality. The main prediction uses raw native g without measured-gain rescaling; the historical pulse-normalized variant is retained in the supplement. Shape encodes f=0.0001,0.0003,0.001,0.003,0.005,0.01 (circle,square,up/down triangle,diamond,plus); size retains the model colour ramp. Horizontal intervals are measured bootstrap 95% intervals. The ratio is approximately the squared edge-gain fold: Wave-pinning $0.70^2 \approx 0.49$; the full spatial covariance gives 0.490, versus 0.473 measured at f=0.001. Shape changes make the square relation approximate, not an exact identity. Absolute predictions and independent-edge candidate are supplementary. Goryachev stays on its own scale and fails quantitatively.

H. Local α=2−dlnD_s/dlnC implied by the predicted drift against Figure 4 weak-input local α (f≤0.001), with identical fixed-neighbour size fits, two neighbours at endpoints and three inside. Grey is equality. Vertical intervals are Figure 4 bootstrap 95% intervals. 14 of 60 requested pairs qualify; Wave-pinning and Goryachev have no qualified weak-input observed α under the fixed shared-lag rule. No missing estimate is replaced. Marker shapes distinguish the three weak inputs as in G.

$$G(x_0) = \sum_j g_j h(x_j - x_0) \tag{3.1}$$

The main map is the standard Gaussian-pulse response per dose, not raw cell-input g.

$$c_j = g_j \sum_k g_k \int_0^\infty \overline{K_u(j,k;t,t+v)}^{\,t}\, \mathrm{d}v, \qquad D_s = \sum_j c_j \tag{3.2}$$

Signed per-position contributions retain the exact native input covariance.

$$\alpha_{\mathrm{pred}} = 2 - \frac{\mathrm{d}\ln D_s}{\mathrm{d}\ln \bar{C}} \tag{3.3}$$

Same fixed adjacent-size neighborhoods as Figure 4; missing observed alpha remains unavailable.